

Kartik Koul

SOFTWARE DEVELOPMENT ENGINEER (SDE) · COMPUTER APPLICATIONS & SCIENCE (MASTERS)

✉ kartikkoulwork@gmail.com

🌐 kartikkoul.com  [github/kartikkoul](https://github.com/kartikkoul)  [linkedin/kartikkoul](https://linkedin.com/kartikkoul)

Skills –

Programming :	Javascript · Typescript · Python · C++/C
Development :	React · Redux · NextJS · NodeJS · ExpressJS · FastAPI · REST APIs · WebSockets
UI Design:	TailwindCSS · FramerMotion · Figma
Data Layer :	PostGres · MongoDB · Mongoose · Prisma · Redis
DevOps:	Docker · AWS · GCP · Monorepo (Turborepo) · CI/CD · Vercel
Tools :	Bash · VS Code · Cursor · Git · GitHub · npm · Postman
AI:	Generative AI · Multimodal · RAG · Agentic AI
AI Frameworks:	LangChain · LangGraph

Experience -

SDE-I , Advanced Voice Research Labs, AVRL

Presently Working

Tech Stack: Javascript, CSS3, HTML5, XPath, Postman, RestAPI

April 2022 – Present

- Automated numerous bots for logistics companies via complex code mechanisms
- Optimized already existing bots & boosted their efficiency to upto 60% where for example an average bot used to take 1 minute to perform certain actions, that dropped down to approximately 25s
- Contributed heavily to various internal software products
- Refined the already existing internal products and reduced the scope of issues in some of them to upto 95%.
- Worked on tickets as well for immediately providing solutions to customer's technical issues

Web Developer , Tech Table (Internship)

7 Months

Tech Stack: Javascript, Python, CSS3, HTML5, React, Redux, Django, Bootstrap, RestAPI

January 2021 – July 2021

- Worked on 3 major projects & 4 minor projects
- Key Speaker in 3 seminars for new batch of interns on Animations, UI Layouts & Web Optimization respectively

Education -

Amity University

Jammu, J&K

Masters in Computer Applications (Majors in Blockchain), 8.2 CGPA

Jul 2022 – Aug 2024

- Amongst top scorers in 4 blockchain development coding contests.
- Took 3 webinars for other students on blockchain technology, ethereum ecosystem & tokenomics respectively.
- MAJOR PROJECT
 - A 30,000 words thesis on “Blockchain in Real Estate”.
 - Developed ERC-720 smart contracts to demonstration tokenization.

University of Jammu

Jammu, J&K

Bachelors in Computer Applications and Science, 7.3 CGPA

August 2018 – October 2021

- As a Student Co-ordinator, managed all kind of activities such as contests, debate or cultural
- All-Time Highest Scorer in practical problem solving exams

- MAJOR PROJECT
 - Introduced modern interface for the first time in our college projects in the final year.
 - Planned out the whole design of the software.
 - Successfully implemented and achieved the modern UI/UX Patterns in our major project which was created using VB.NET as Front End and Backend and SQL Server as the database

Projects —

My KnowledgeBase

Tech Stack

<https://github.com/kartikkoul/rag-pdf-qna>

NextJS | FastAPI | PostgreSQL | Prisma | Langchain | Docker | AWS | Vercel

My Knowledgebase is a full-stack Retrieval-Augmented Generation (RAG) application that lets users upload PDF documents to build a private, personal knowledge base, then ask questions that are answered strictly from the uploaded content with token-by-token streaming, hybrid retrieval, and per-user data isolation.

My KnowledgeBase

Tech Stack

<https://github.com/kartikkoul/sketchmates>

NextJS | NodeJS | ExpressJS | WebSockets | PostgreSQL | Prisma | Turborepo | AWS

Sketchmates lets you draw whatever you imagine: together, in real time. Perfect for just goofing around with friends..